Observer Patch Holography and the Dark Sector

Modular Charge, the Anomalous Collar Source, and the Deep Galaxy Law

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Abstract

Under a supplied universal coupling, geometry responds to modular charge without regard to electromagnetic visibility. An overlap-collar recovery defect provides a conditional non-luminous source whose stress is bounded by that defect and vanishes at full recovery. A common exponential recovery envelope suppresses compact sources; an erasure-record model supplies an explicit defect law. For nonnegative anomalous mass, deep-regime scale covariance and quadrature composition uniquely give $a_A = \sqrt{a_b a_0}$ and $v^4 = GM_b a_0$, with one undetermined constant. A per-cut model expresses it as $a_0 = Gn^2 c$. These are conditional source and profile results, not a derivation of the physical cut density, variance or relativistic stress. Retrospective galaxy fits test the profile and reveal material weighting and covariance sensitivity. Lensing, cosmological abundance and perturbation dynamics require a common covariant completion.

1 What Sources Gravity in OPH

The finite classification, recovery and profile identities are checked under the stated hypotheses in the Lean formalization [15]. Their physical identification is an additional premise throughout.

The conditional Einstein arrow of the spacetime paper ends in

$$G_{ab} + \Lambda g_{ab} = 8\pi G \langle T_{ab} \rangle, \tag{1}$$

where $\langle T_{ab} \rangle$ is a supplied local, covariant, conserved stress on the physical branch [1]. Its smooth form requires same-family tensor-curvature convergence or the continuum small-ball/null-balance identification; scalar-curvature convergence alone is diagnostic. Finite positive-null tomography exactly reconstructs a quadratic form from directional charges, but identifying that form with the same local stress additionally requires charge–stress matching, locality and covariance, and a weak Ward law. In the rest frame the relation then follows from three further premises on a small geodesic ball of radius ℓ: generalized entropy is declared stationary, the bulk entropy variation equals $(8\pi^2\ell^4/15)\,\delta\langle T_{00}\rangle$, and the fixed-volume area variation equals $-(4\pi\ell^4/15)\,\delta G_{00}$. Exact arithmetic gives $\delta G_{00} = 8\pi G\,\delta\langle T_{00}\rangle$.

On that conditional branch, the stress entering this relation is identified with the same modular charge. The screen does not consult the electromagnetic readout of a source before responding to it. That is the universal-coupling premise, and it is the only input the dark-sector question takes from the gravity derivation.

2 The Generic Dark Sector

Let a finite family of screen sources be indexed by i, each with rest-frame modular-charge contribution q_i and a flag recording whether it carries the electromagnetic readout. Write Q_{lum} for the sum over flagged sources and Q_{dark} for the sum over the others, so that $Q_{\mathrm{tot}} = Q_{\mathrm{lum}} + Q_{\mathrm{dark}}$.

Premise 2.1 (Universal coupling)**.** *The stress contraction entering the rest-frame Einstein relation equals Q_{tot}, with no readout-dependent weight.*

Proposition 2.2 (Total and non-luminous response)**.** *Under Premise 2.1 and $G > 0$, $\delta G_{00} = 8\pi G(Q_{\mathrm{lum}} + Q_{\mathrm{dark}})$. The luminous-only equation holds exactly when $Q_{\mathrm{dark}} = 0$. If every non-luminous charge is nonnegative, its weak-field contribution has the attractive sign.*

Proof. Substitute the finite-sum decomposition of Q_{tot} in the assumed response law. Cancel $8\pi G$ for the equivalence and sum nonnegative terms for the sign. □

This classifies a geometric excess if one exists; it does not establish a nonzero charge, its abundance, or a physical carrier.

3 The Modular-Anomaly Source

On the support-visible branch the cap modular generator has the local form

$$K_C = 2\pi B_C + K_C^{\mathrm{anom}} + \mathrm{const}, \tag{2}$$

with B_C the bulk boost generator and K_C^{anom} the nonadditive part carried on the overlap collar [1]. At finite regulator stage the separately supplied collar-recovery model carries this part as a remainder and bounds it:

$$\left|\delta\langle E_C^{(\delta)}\rangle\right| \le C_C\,\eta_\delta, \tag{3}$$

where η_δ is the collar recovery defect and C_C depends only on the collar model and the bounded observable class.

Definition 3.1 (Anomalous stress and density)**.**

$$\delta\langle T_{00}^{\mathrm{anom}}\rangle := \frac{15}{8\pi^2\ell^4}\,\delta\langle E_C^{(\delta)}\rangle, \qquad \rho_A := \frac{1}{c^2}\,\delta\langle T_{00}^{\mathrm{anom}}\rangle. \tag{4}$$

The dark-sector adapter identifies this anomalous modular energy with the non-luminous charge of Section 2. Its absence of an electromagnetic readout is a source-localization hypothesis: it is assigned to the part of the modular generator that no local record exposes. Neither identification follows from the finite Einstein-shape theorem. Three consequences follow from (3) and (4).

Theorem 3.2 (Bounded source)**.** $\left|\delta\langle T_{00}^{\mathrm{anom}}\rangle\right| \le 15\,C_C\,\eta_\delta/(8\pi^2\ell^4)$.

Theorem 3.3 (Full recovery switches the source off)**.** *If $\eta_\delta = 0$ then $\delta\langle T_{00}^{\mathrm{anom}}\rangle = 0$. Along any family with fixed ℓ, bounded C_C, and $\eta_\delta \to 0$, the anomalous stress tends to zero.*

Theorem 3.4 (Cold scaling)**.** *A conserved anomalous charge in a comoving cell of physical volume $\propto a^3$ has density $\propto a^{-3}$.*

Proofs. Theorem 3.2 multiplies (3) by the positive kernel inverse. Theorem 3.3 is the case $\eta_\delta = 0$ of the bound and a squeeze along the family. Theorem 3.4 is $\rho\, a^3 = Q$. □

The split (2) has a machine-checked finite counterpart. On a finite carrier of microstates labelled bulk or collar, with strictly positive weights, a declared boost datum vanishing on the collar, and modular generator $K = -\log w$, the decomposition $K = 2\pi B + K^{\mathrm{anom}} + c\,\mathbf{1}$ exists with c the bulk-weighted mean of $K - 2\pi B$, is unique among splits whose anomalous part has vanishing bulk mean, and is invariant under the additive-constant quotient $w \mapsto e^{-t} w$. When the bulk weights are exactly Gibbs for $2\pi B$, the anomalous part vanishes on the bulk and is carried entirely on the collar, and its expectation is the remainder bounded in (3). The finite carrier, the boost datum, the normalization clause, and the Gibbs condition are model premises; the receipt is the finite counterpart of (2), at exactly that strength. The executable observer does not supply these premises: its full configuration reference approaches the normalized trace at fixed modular parameter, so it identifies no quantum vacuum and its tracial limit has trivial modular flow, and a bulk/collar split of its source is not constructed [11, 12]. Its regional bulk/edge entropy split on one seam carries no boost generator or geometric normalization [1].

In the weak-field limit the anomalous density enters the Poisson equation beside the baryons,

$$\nabla^2 \Phi = 4\pi G\,(\rho_b + \rho_A), \tag{5}$$

with the attractive sign of Proposition 2.2 whenever the anomalous charge is nonnegative.

4 Galactic-Scale Collars and the Solar System

The collars in (2) are overlap collars between observer patches, and the recovery defect η_δ measures how far the repaired record on a collar falls short of determining its complement. Where records are dense and repair saturates, the defect is small; where the screen is sparsely recorded, over the settled outskirts of a galaxy, the defect is carried on long-lived cuts. The consensus-layer provenance package supplies a narrower finite statement [2]: an injected overlap mismatch persists at full strength through any continuation in which no later commit writes a register in that overlap support. This is an unwritten-support persistence theorem. It does not show that a physical galactic cut satisfies the no-write hypothesis or remains long-lived. The proposed physical identification assigns this anomalous record state to galactic collars. The native continuum of one repair history is a two-dimensional spatial operator on the screen with no radial domain, and its internal rank-three modes vanish from bounded-energy families; a bulk reading needs a radial observation algebra that the native energy does not supply [10]. The map from screen collars to nested three-dimensional galactic shells is therefore part of this identification.

Premise 4.1 (Collar scale)**.** *The collars carrying $\delta\langle E_C^{(\delta)}\rangle$ are galactic-scale settled cuts. A compact source inside the galaxy does not generate anomalous charge of its own; it sits in the ambient anomalous density ρ_A of its neighbourhood.*

A conditional saturation theorem supports limiting suppression in the compact-source part of this premise. The collar-recovery claim of the spacetime paper [1] bounds the defect on its conditional mixing branch by an exponential envelope, which we write in the local record density ρ as

$$\eta_\delta \le \kappa\, B\, \exp\!\Big(-\frac{\delta \rho^{1/3} - r_0}{\zeta}\Big), \tag{6}$$

with κ the mixing prefactor, B the boundary size in UV cells, r_0 an offset, ζ the decay constant, and $\delta\rho^{1/3}$ the collar depth counted in UV record spacings.

Theorem 4.2 (Compact-source saturation)**.** *Under the envelope (6), fix common parameters $\kappa \geq 0$, $B \geq 1$, $r_0 \geq 0$, and $\zeta > 0$. Along a family of collars with fixed radius $\ell > 0$ and uniformly bounded nonnegative collar constant C_C, the recovery defect and anomalous stress tend to zero as the depth tends to infinity at fixed positive record density, or as the record density tends to infinity at fixed positive depth. Quantitatively,*

$$|\delta\langle T_{00}^{\mathrm{anom}}\rangle| \leq \frac{15\, C_C}{8\pi^2\ell^4}\,\kappa B\,\exp\Big(-\frac{\delta\rho^{1/3} - r_0}{\zeta}\Big), \tag{7}$$

For $\kappa B > 0$ and $\varepsilon > 0$, the defect envelope bound falls below ε exactly when $\delta\rho^{1/3} > r_0 + \zeta \log(\kappa B/\varepsilon)$. For $\kappa = 0$, the defect and stress vanish at every admissible depth. Thus the bounded stress tends to zero in these fixed-parameter limits.

The delimitation is exact in both directions and machine-checked. The identically zero defect satisfies every envelope hypothesis, so the envelope never forces a nonzero defect: that the galactic-scale settled cuts carry an order-one defect is a premise, not a consequence. And an explicit sparsely recorded instance sits at the bound with an order-one defect and strictly positive stress, so the same hypotheses that screen dense regions admit a nonzero dark source on sparse cuts. What is not established is that physical collars satisfy (6): the envelope is the conditional mixing branch of the collar-recovery claim, taken here as a hypothesis. The executable observer's native repairs have an exact quantum mixing gap $(5 - \sqrt{5})/60$ per uniform seam attempt on the twelve-port carrier [13]; that gap concerns relaxation in native event count and supplies no envelope in collar depth.

The delimitation has a positive counterpart in which the defect is computed rather than admitted. In the erasure-record model, a source bit perfectly correlated with its complement and a collar record carrying the bit with probability p, the minimum prediction error over all predictors, deterministic or randomized, is exactly $(1 - p)/2$: an unrecorded cut carries an order-one defect necessarily, a saturated record carries none, and the defect falls below ε exactly when $p > 1 - 2\varepsilon$, the mechanism-side counterpart of the settled-cut threshold. With unit collar radius and collar constant, and anomalous energy equal to the model defect, the unrecorded instance carries stress $15/(16\pi^2)$ exactly. The identification of the model defect with η_δ is a modeling premise, and the record density of a physical galactic-scale cut is not specified; the model shows the mechanism class exists, with an exact defect-density law.

Under Premise 4.1 the Solar System is a test of the ambient density alone.

Proposition 4.3 (An isotropic ambient density has no quadrupole)**.** *The potential of a locally uniform density is $(2\pi G\rho_A/3)\, r^2$. Its projection on the quadrupole weight $P_2(\cos\theta)$ vanishes, $\int_{-1}^{1}(\mu^2 - \frac{1}{3})\,\mathrm{d}\mu = 0$, so the quadrupole coefficient Q_2 is zero. The only footprint is the isotropic tidal scale $4\pi G\rho_A/3$.*

With the local dark density $\rho_A \simeq 6.8\times10^{-22}\,\mathrm{kg\,m^{-3}}$, the value inferred from Galactic kinematics [9], the tidal scale is

$$\frac{4\pi G\rho_A}{3} = 1.9\times10^{-31}\,\mathrm{s^{-2}}, \tag{8}$$

For a dimensional scale comparison, the Cassini trace-free quadrupole result is $Q_2 = (1.6 \pm 1.8)\times 10^{-27}\,\mathrm{s^{-2}}$ [8], four orders of magnitude above the isotropic scale. It is not a direct bound on the

traceful isotropic channel, whose quadrupole is identically zero. The anomalous mass enclosed within ten astronomical units is $5 \times 10^{-15}\, M_\odot$. Both numbers are machine-checked inequalities in the formalization. The Solar System constrains the premise of Section 4 and nothing else in the construction.

5 The Deep-Regime Profile

Around a baryonic mass M_b, the proposed collar-count interpretation assigns a number of settled cuts proportional to the enclosing radius r. The precise deep-regime characterization uses the following two premises, each independently falsifiable.

Premise 5.1 (Deep-regime scale covariance)**.** *In the deep regime the enclosed anomalous mass is degree-one homogeneous in the radius: $M_A(M_b, \lambda r) = \lambda\, M_A(M_b, r)$ for every $\lambda > 0$. A sparsely recorded deep cut retains no length scale, so rescaling the radius rescales the enclosed mass by the same factor.*

Premise 5.2 (Quadrature composition)**.** *The anomalous amplitudes of independent baryonic sources compose in quadrature: $M_A(m_1 + m_2, r)^2 = M_A(m_1, r)^2 + M_A(m_2, r)^2$. The collar defects of distinct sources are independent, their variances add, and the gravitating anomalous charge is the amplitude of the unrecovered record fluctuation.*

Theorem 5.3 (Profile characterization)**.** *Let $M_A(M_b, r)$ be nonnegative, monotone in the source, somewhere nonzero, and satisfy Premises 5.1 and 5.2. Then there is a unique constant $a_0 > 0$ with*

$$M_A(r) = r\sqrt{\frac{M_b a_0}{G}}, \qquad \rho_A(r) = \frac{1}{4\pi r^2}\sqrt{\frac{M_b a_0}{G}}. \tag{9}$$

Proof. Scale covariance factorizes $M_A(M_b, r) = r\, f(M_b)$. Quadrature composition makes f^2 additive on positive sources, monotonicity makes it monotone, and a function additive and monotone on the positive reals is linear: $f^2 = \lambda M_b$ with $\lambda > 0$ by nondegeneracy. Setting $a_0 = \lambda G$ gives (9); uniqueness follows by evaluating at $M_b = r = 1$. Each premise is load-bearing: an additive law $M_A \propto r M_b$ satisfies everything except quadrature composition, and an r^2 law everything except scale covariance. □

Corollary 5.4 (Linear enclosed mass)**.** *The deep-regime density is the shell derivative of the enclosed mass, $M_A'(r) = 4\pi r^2 \rho_A(r)$, with M_A linear in r and the coefficient $\sqrt{M_b a_0/G}$ forced by Theorem 5.3.*

Deep-limit scale invariance is the known symmetry of the modified-dynamics limit [6], and square-root mass laws appear in entropic-gravity proposals [7]. The resulting law is the deep limit of modified Newtonian dynamics, $a_A = \sqrt{a_b a_0}$, with its baryonic Tully–Fisher corollary; the characterization adds that the two premises force it jointly, and it reads the square root as the variance additivity of independent collar defects under source composition. It does not fix a_0.

Premise 5.2 itself follows from a more primitive per-cut model. Suppose the nr nested cuts within radius r each carry anomalous amplitude $\sqrt{V(M_b)}$, the standard deviation of the collar defect sourced by the baryons, with the variances of independent sources adding and the variance monotone in the source. Then the variance function is forced linear, $V(M_b) = cM_b$ with a unique $c > 0$,

quadrature composition is a theorem rather than a premise, the induced law satisfies Theorem 5.3, and its constant is exactly

$$a_0 = G\, n^2 c. \tag{10}$$

The dictionary is exact but determines only the product $n^2 c$: two models with different cut densities and equal $n^2 c$ induce the same enclosed-mass law, machine-checked. The magnitude question is therefore the value of the collar density and of the per-cut variance constant; neither is derived here.

One sufficient realization of both premises rests one level deeper. Variance additivity follows from probabilistic independence: for atomic defect contributions with finite second moments, pairwise independence, and a common per-atom variance on one probability space, the variance of the compound defect of every finite atom family is the sum of the per-atom variances, so disjoint-union composition is additive and $V(m) = cm$ on the modeled masses. Independence is stronger than necessary; pairwise zero covariance and other dependent but uncorrelated constructions are viable. An exact linear-asymptotic cut count follows from one rigid geometry: a cut family with a positive first cut, consecutive gaps of at least s, and every length-s interval meeting a cut is exactly the arithmetic progression with spacing s, its count obeys $r/s - 1 \le N(r) \le r/s + 1$, and the counted enclosed mass deviates from the linear law by at most one per-cut amplitude at every radius, with $n = 1/s$. A perfectly correlated pair and a one-cut family show only that the conclusions do not follow after deleting every decorrelation or density condition. They do not make independence or exact spacing necessary. Irregular or ergodic cut families with asymptotic density and sublinear count error are possible. The exact nr per-cut law is the continuum or mean extension; the discrete count above differs from it by a bounded error.

Theorem 5.5 (Deep radial-acceleration relation)**.** *The anomalous acceleration of the profile* (9) *is*

$$a_A(r) = \frac{GM_A(r)}{r^2} = \frac{\sqrt{GM_b a_0}}{r} = \sqrt{a_b(r)\, a_0}, \qquad a_b = \frac{GM_b}{r^2}. \tag{11}$$

Theorem 5.6 (Flat rotation curve and baryonic Tully–Fisher relation)**.** *The circular speed supported by a_A satisfies $v^2 = a_A r = \sqrt{GM_b a_0}$, independent of r, and hence*

$$v^4 = GM_b\, a_0. \tag{12}$$

Proofs. $GM_A/r^2 = G\sqrt{M_b a_0/G}/r = \sqrt{GM_b a_0}/r$, and $\sqrt{GM_b a_0}/r = \sqrt{(GM_b/r^2)a_0}$ for $r > 0$. Multiplying by r removes the radius; squaring gives (12). □

The exponents 1/2 in (11) and 4 in (12) are the parameter-free content of the profile. The constant a_0 collects C_C, the collar density, and the small-ball normalization; none of these is fixed by the source, so a_0 enters every comparison as a fitted value. The deep regime is the range $a_b \ll a_0$, equivalently $r \gg r_M = \sqrt{GM_b/a_0}$. If the deep expression is algebraically extended to the matching radius, both accelerations equal a_0 there and the anomalous term dominates outside r_M; those algebraic statements are machine-checked. The premises are asserted only in the deep regime, so they do not determine the physical transition or the response inside r_M.

6 Lensing and the Relativistic Source

The anomalous charge must enter (1) through one stress tensor; Premise 2.1 supplies no readout-dependent coupling. That fact does not imply equality of masses inferred from null geodesics and

timelike orbits. Those observables contract different components of the metric response and can differ in the presence of pressure, anisotropic stress, or gravitational slip. No relativistic refinement limit is supplied that turns the weak-field density (9) into a full T_{ab}^{anom}. A lensing law and the relation between lensing and dynamical mass are therefore physical constructions that this paper does not supply.

7 Comparison with Rotation-Curve Data

Theorems 5.5 and 5.6 are compared with the SPARC sample of 175 disk galaxies with 3.6 μm photometry [4] under the declared inclusive-error deep-law diagnostic: quality flag at most 2, inclination at least 30°, velocity error at most ten percent, and stellar mass-to-light ratios 0.5 for disks and 0.7 for bulges. This leaves 2700 points in 149 galaxies. The strict published-cut replay described below is kept separate. Both computations are released with the paper [5, 3].

Source-bound full-RAR calibration. Separate from the conditional OPH deep law, a deterministic replay fits the published empirical interpolation

$$g_{\mathrm{obs}} = \frac{g_{\mathrm{bar}}}{1 - \exp[-\sqrt{g_{\mathrm{bar}}/a_0}]}$$

to a committed CDS/VizieR snapshot whose provenance, catalogue description, and two table files are pinned by SHA-256. The catalogue cuts reproduce the published 153-galaxy parent selection. The strict point cuts retain 2696 points in 147 contributing galaxies, three more than the published 2693-point census. These counts are distinct from the preceding 2700-point, 149-galaxy deep-law diagnostic: that diagnostic uses the inclusive cut $dV/V \leq 0.1$, whereas this full-RAR replay uses the strict cut $dV/V < 0.1$ and starts from the 153-galaxy parent set. With disk and bulge mass-to-light values 0.5 and 0.7, the unweighted point-level log-residual fit gives

$$a_0 = (1.1613 \pm 0.0802) \times 10^{-10}\,\mathrm{m\,s^{-2}}, \qquad \sigma_{\log_{10} g} = 0.1327 \text{ dex},$$

where the width is the standard deviation of a galaxy bootstrap. Equal total weight per contributing galaxy shifts a_0 by -9.58%; weighting only by inverse fractional velocity variance shifts it by $+15.49\%$. These are estimator-sensitivity controls, not calibrated likelihood alternatives. The interpolation and acceleration scale were known before this replay, no OPH source value enters the fit, and distance, inclination, gas, stellar population, intrinsic scatter, and correlation uncertainties are not jointly marginalized. The result certifies public-data custody and arithmetic, not independent evidence for OPH [5, 3].

Radial acceleration. On fixed absolute subsets $g_{\mathrm{bar}} < f a_{\mathrm{ref}}$, with the comparison-only value $a_{\mathrm{ref}} = 1.2 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ fixed before fitting, the density profile predicts the total acceleration $g_{\mathrm{obs}} = g_{\mathrm{bar}} + \sqrt{g_{\mathrm{bar}} a_0}$. The fixed-exponent diagnostic gives

$$a_0 = 0.843 \times 10^{-10}, \quad 0.875 \times 10^{-10}, \quad 0.924 \times 10^{-10}\ \mathrm{m\,s^{-2}}$$

for $f = 0.3$, 0.1, and 0.03, with 1519, 960, and 203 points in 141, 115, and 40 galaxies and rms scatter between 0.14 and 0.17 dex. Fitting an anomalous power inside the total model, using every retained point rather than selecting on the sign of the noisy excess, gives exponents 0.454, 0.461,

and 0.239. Galaxy-cluster bootstrap intervals contain 1/2 for the first two cuts; the deepest cut is sparse and its interval is wide. The computation uses equal retained-point weighting and does not marginalize distance, inclination, stellar-population, or correlated rotation-curve uncertainties, so it is a diagnostic rather than a full measurement likelihood.

Tully–Fisher and matched channels. The 123 quality-selected galaxies with positive flat velocity give $a_0 = v_f^4/(GM_b) = 1.55 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, with 0.26 dex scatter, for $M_b = 0.5L_{[3.6]} + 1.33M_{\mathrm{HI}}$. The ordinary least-squares exponent is 3.75 ± 0.10, placing the fixed exponent 4 at $+2.66\sigma$ in this diagnostic. Catalogued flat speeds retain a baryonic contribution at finite radius and need not measure the anomalous speed entering the theorem.

A matched comparison uses $v_A^2 = v_{\mathrm{obs}}^2 - v_{\mathrm{bar}}^2$ at the outermost radius, $g_{\mathrm{bar}} < 0.1a_{\mathrm{ref}}$, and excludes nonzero bulge components without separate luminosities. On the resulting 97-galaxy set the speed and radial channels give respectively 0.769 and $0.853 \times 10^{-10}\,\mathrm{m\,s^{-2}}$. Their paired log-ratio is -0.045 dex with 95 percent galaxy-bootstrap interval $[-0.117, +0.026]$. For comparison, the mixed flat-speed/radial analysis has interval $[0.180, 0.317]$ dex and normalization intervals $[0.730, 1.037]$ and $[1.405, 1.723] \times 10^{-10}\,\mathrm{m\,s^{-2}}$. The matched construction changes radius, subtraction, sample and combination together; it does not isolate one cause for the discrepancy. It is consistent with one fitted constant on seen data, not a physical likelihood verdict.

Covariance and nuisance sensitivity. A separate penalized Gaussian objective profiles disk mass-to-light, distance and inclination, and scans an equal-correlation family $\rho \in \{0, 0.2, 0.4, 0.6\}$. A log-normal mass-to-light prior centered at 0.5 is a penalty rather than an auxiliary-data likelihood. The fixed catalogue-frame mean is $v_{\mathrm{model}} \sin i / \sin i_{\mathrm{cat}}$. Transforming observations and errors by $s(i) = \sin i_{\mathrm{cat}} / \sin i$ also requires the density Jacobian $s(i)^n$, which cancels the Gaussian factor $s(i)^{-n}$.

On the $f = 0.1$ subset the optimum shifts from 0.804 to $0.582 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, or 0.140 dex, as ρ increases from zero to 0.6. The data quadratic form per nominal fixed-nuisance degree of freedom lies between 3.6 and 16.4; the family has no intrinsic scatter and understates observed variation. Its threshold sets are uncalibrated sensitivity contours, not confidence intervals. The paired comparison contains zero at every scanned setting, but channel-B nuisances are held at full-sample estimates while channel A is refitted, so it is only partially paired. Independent replay and synthetic controls verify the subtraction, fixed observation frame, covariance algebra and component enumeration [3]. A physical test requires a calibrated covariance and intrinsic-scatter likelihood.

8 Cosmological Scaling

If the total anomalous charge in a comoving region is conserved, Theorem 3.4 gives $\rho_A \propto a^{-3}$: the anomalous medium has the homogeneous dilution law of a pressureless component. On the sourced causal order of the finite-source cosmology paper the same statement reads: the layered read law in comoving coordinates is the flat order for every scale-factor profile, a comoving cell at layer j has physical volume a_j^3 times its comoving volume, and a layer-independent anomalous charge per comoving cell is the density $\rho_A \propto a_j^{-3}$ by definition of that count measure. Conservation of the charge under comoving strain is the operational input: fixed-volume records of the same Q do not select the exponent $\alpha = 0$ of $H_\alpha(V) = Q(V/V_0)^{-\alpha}$, for which $p/\rho = \alpha$ [14]. This does not derive

its pressure perturbations, anisotropic stress, sound speed, or clustering. Premise 4.1 associates its carrier with galactic collars but supplies no perturbation dynamics. The abundance Ω_A, initial data, transfer functions, and the claim that it clusters with galaxies are not derived here.

9 Physical closure conditions

The source interpretation requires the same cap modular generator, record recovery law, local stress, physical clock and geometric response. The finite bulk/collar split supplies the anomaly under its Gibbs-on-bulk premise; it does not identify a physical boost generator. The common recovery envelope suppresses compact sources only in its stated fixed-parameter limit and does not force a nonzero galactic defect. The physical location, density and statistics of those cuts must therefore be supplied separately.

The profile theorem determines the exponents and leaves $a_0 = Gn^2c$. A constant-capacity horizon dictionary supplies a separate dimensional scale,

$$a_{\mathrm{dS}} = c^2\sqrt{\Lambda/3} = \frac{c^2}{\ell_P}\sqrt{\frac{\pi}{N}}. \tag{13}$$

At the Planck-central $\Lambda = 1.089 \times 10^{-52}\,\mathrm{m}^{-2}$, this is $5.42 \times 10^{-10}\,\mathrm{m\,s}^{-2}$. The full-RAR calibration gives $a_0/a_{\mathrm{dS}} = 0.214$, ranging from 0.176 to 0.268 as disk mass-to-light varies from 0.6 to 0.4; the deep-law fits give 0.156 to 0.171. These comparisons do not select n or the per-cut variance constant. In this paragraph the c in a_{dS} is the speed of light, distinct from the variance constant in the per-cut dictionary.

A covariant completion must determine pressure, anisotropic stress, gravitational slip, exchange, abundance and perturbations before lensing, clusters or cosmology can test the same source. The static deep law neither specifies the transition to Newtonian dynamics nor establishes those quantities. Its galaxy comparisons are retrospective fits with declared cuts and nuisances; a source-derived acceleration requires an independent physical scale and a prospectively fixed measurement likelihood.

References

[1] B. Mueller et al., *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency*, OPH spacetime and Einstein paper. https://github.com/FloatingPragma/observer-patch-holography/blob/r2061/paper/recovering_observer_spacetime_and_einstein_dynamics_from_overlap_consistency.pdf

[2] B. Mueller et al., *Reality as a Consensus Protocol*, OPH consensus paper. https://github.com/FloatingPragma/observer-patch-holography/blob/r2061/paper/reality_as_consensus_protocol.pdf

[3] B. Mueller, *Deep-regime comparison of the OPH galaxy law with SPARC*, 2026. https://github.com/FloatingPragma/observer-patch-holography/tree/r2061/code/cosmology/rar_deep_regime

[4] F. Lelli, S. S. McGaugh, and J. M. Schombert, *SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves*, Astron. J. 152, 157 (2016),

arXiv:1606.09251. https://arxiv.org/abs/1606.09251. CDS/VizieR catalogue: https://cdsarc.cds.unistra.fr/ftp/J/AJ/152/157/

[5] S. S. McGaugh, F. Lelli, and J. M. Schombert, *Radial Acceleration Relation in Rotationally Supported Galaxies*, Phys. Rev. Lett. 117, 201101 (2016), arXiv:1609.05917. https://arxiv.org/abs/1609.05917

[6] M. Milgrom, *The MOND limit from spacetime scale invariance*, Astrophys. J. 698, 1630 (2009), arXiv:0810.4065. https://arxiv.org/abs/0810.4065

[7] E. Verlinde, *Emergent Gravity and the Dark Universe*, SciPost Physics **2**, article 016 (2017), arXiv:1611.02269. https://arxiv.org/abs/1611.02269

[8] R. S. Park et al., *Improved constraints on modified Newtonian gravity from Cassini radio tracking data*, 2026, arXiv:2602.17884. https://arxiv.org/abs/2602.17884

[9] P. F. de Salas and A. Widmark, *Dark matter local density determination*, Rep. Prog. Phys. 84, 104901 (2021), arXiv:2012.11477. https://arxiv.org/abs/2012.11477

[10] FloatingPragma (2026). A spatial fluctuation field from one native history: the common spatial generator, its locality and spectral dimension, and the boundary-trace test. https://github.com/FloatingPragma/observer-patch-holography/blob/r2061/model/physical_space_time/OCCUPATION_FIELD.md

[11] FloatingPragma (2026). Joint time and field laws on matched source screens. https://github.com/FloatingPragma/observer-patch-holography/blob/r2061/model/observer_time_fields/REFINEMENT.md

[12] FloatingPragma (2026). Tangent first laws and finite entropy changes: native reference entropy and region boundaries. https://github.com/FloatingPragma/observer-patch-holography/blob/r2061/code/geometry/ENTROPY_FIRST_LAW_AUDIT.md

[13] FloatingPragma (2026). Native quantum mixing and the full-carrier memory boundary. https://github.com/FloatingPragma/observer-patch-holography/blob/r2061/model/observer/QUANTUM_MIXING.md

[14] B. Müller et al., *Observers Are All You Need*. https://github.com/FloatingPragma/observer-patch-holography/blob/r2061/paper/observers_are_all_you_need.pdf.

[15] B. Mueller et al., *Observer Patch Holography: machine-checked Lean formalization*, Einstein-branch modules `SmallBall`, `DarkSector`, `CollarScaleSaturation`, `DeepProfileClosure`, `PerCutCollarComposition`, `SparseRecordDefectWitness`, `CollarPremiseDerivation`, and `FiniteCapGeneratorSplit`, 2026. https://github.com/FloatingPragma/observer-patch-holography/tree/r2061/Lean/ObserverPatchHolography/EinsteinBranch

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